

CE 310

Excel Tutorial — Histograms, Skewness, and Percentiles

Week 2 · Session 2 · Hands-on Walkthrough

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Today's Plan — Four New Skills, Then the Lab

By the end of this walkthrough you will know how to:

1. Count observations into histogram bins using **COUNTIFS** with two conditions
2. Insert a **histogram chart** with bars that touch (no gap)
3. Compute **SKEW()** and read the result
4. Insert a **box-and-whisker chart** directly from raw data
5. Use **PERCENTILE()** to find any tail value in one formula

Assumed from Week 1: you can open the CSV, freeze the header, and write `=AVERAGE` , `=STDEV` , `=COUNTIF` . Today builds on top of those.

Step 0 — Bring Your Data Across

Your Week 2 Answer Sheet is a **new file**. It ships with `Instructions`, `Answers`, `Work`, `Charts` and `Written` — and **no Data sheet**. Every formula today reads `Data!`, so build it before anything else.

 **Fastest — copy it from Week 1.** Open your Week 1 file, right-click the `Data` tab → **Move or Copy** → *To book:* `Wk02.04_In_Class_Answer_Sheet.xlsx` → tick **Create a copy** → **OK**. The `Month` helper column comes across with it.

 **If Excel warns about connections or links**, don't fight it. Select `A1:E8761` on your Week 1 `Data` sheet → **Copy** → on a new sheet named `Data` here, click `A1` → **Paste Special** → **Values**. Or rebuild it exactly as in Week 1: **Data** → **Get Data** → **From Text/CSV**, loaded to `$A$1`.

Step 0 — Confirm Before You Go On

Header in row 1, data in rows **2 to 8761** — 8,760 hours.

A	B	C	D	E
DateTime	Temp_F	Wind_Speed_mph	Relative_Humidity_pct	Month

 **If column E is empty**, rebuild it: type `Month` in `E1`, `=MONTH(A2)` in `E2`, and fill down to row 8761.
Section B's humidity questions read this column.

 The sheet must be named exactly `Data`. If a formula returns `#REF!` or Excel opens a file-picker, that name is what to check first.

Step 1 — Find the Bin Table

A histogram needs a table: one column of **bin labels**, one column of **counts**. Your Answer Sheet ships with it laid out already, so you build it once and the lab reads it from there.

 **Open Wk02.04_In_Class_Answer_Sheet.xlsx and click the Work tab.** Find the first block, *Questions A1-A3 + Chart 1*:

Column A — Bin low · each bin's lower edge, as a **number**: 20, 30, 40 ... 110

Column B — Bin (°F) · the label, 20–30 through 110–120

Column C — Hours · yellow and empty. This is the column you fill.

Ten bins, 20–120°F in 10°F steps — wide enough for Tucson's full 28–112°F range. Column A is a **number**, not text, because your formula does arithmetic on it.

Data! is not optional. The table is on **Work**; the readings are on **Data**. Every range you write outside the **Data** sheet needs that prefix — leave it off and Excel searches the sheet you are standing on, finds nothing, and returns 0.

Step 2 — COUNTIFS with Two Conditions

To count hours in a bin, you need **two simultaneous conditions** on the same column: a lower bound AND an upper bound.

& joins a text string to a cell's value. `">="&A7` becomes `">=20"` in row 7 and `">=30"` in row 8. You cannot write `">=A7"` — Excel treats that as literal text.

\$ locks a row number when filling down. In `Data!B$2:B$8761` the `$` before 2 and 8761 keeps those rows fixed; without it, filling down shifts to `B3:B8762`. `A7` (no `$`) advances to A8, A9...

 **Click cell C7 and type:**

```
=COUNTIFS(Data!B$2:B$8761,">="&A7,Data!B$2:B$8761,"<"&(A7+10))
```

Press Enter. You should see **6** — very few hours below 30°F in Tucson.

Step 2 — Building the Bin Table

Step 2 — what the & signs actually build

C7

fx

=COUNTIFS(Data!B\$2:B\$8761,">="&A7,Data!B\$2:B\$8761,"<"&(A7+10))

	A	B	C
6	Bin low	Bin (°F)	Hours
7	20	20-30	6
8	30	30-40	280
9	40	40-50	771
:			
16	110	110-120	9

& glues text onto a cell's value. You cannot write ">=A7" — Excel would read that as the literal text "A7".

row 7: ">="&A7 → ">=20" "<"&(A7+10) → "<30"

row 8: ">="&A8 → ">=30" "<"&(A8+10) → "<40"

row 9: ">="&A9 → ">=40" "<"&(A9+10) → "<50"

Each row builds its own pair of bounds from the number sitting in column A — which is why the bin table works at all.

Ten bins, 20-120 °F. The tallest is 80-90 °F at 1,598 hours — the modal bin.

Column A holds the numbers. The formula builds its own pair of text bounds from them, one row at a time.

Step 3 — Fill the Bin Table Down

 **Select C7. Double-click the fill handle** (bottom-right corner of the cell) to fill C8 through C16 automatically.

Check these four before moving on:

Cell	Bin	Expected
C7	20–30°F	6
C10	50–60°F	≈ 1,358
C13	80–90°F	≈ 1,598 ← modal
C16	110–120°F	≈ 9

 **The wind table below it is not a paste of this one.** Those bins are **5 mph wide, not 10**: the upper bound becomes `"<"&(A21+5)` , and the column changes from `Data!B` to `Data!C` . Paste the temperature formula unchanged and the 0–5 bin reads about 6,700 instead of 3,038.

Step 3 — The Fill-Down Trap

Why the \$ signs — and why this bug hides

With \$ — Data!B\$2:B\$8761

C7: =COUNTIFS(Data!B\$2:B\$8761,">="&A7, ...)

C16: =COUNTIFS(Data!B\$2:B\$8761,">="&A16,...)

The range never moves. Only the bin boundary advances, which is the whole point of filling down.

Without \$ — Data!B2:B8761

C7: =COUNTIFS(Data!B2:B8761,">="&A7, ...)

C16: =COUNTIFS(Data!B11:B8770,">="&A16,...)

The range slides one row per fill. By C16 it has dropped the first nine hours and run nine rows past the data.

	A	C	D
6	Bin low	with \$	no \$
10	50	1,358	1,358
11	60	1,506	1,503
12	70	1,583	1,583

⚠️ Nine bins agree exactly.

Only the 60–70 °F bin moves — 1,506 becomes 1,503. That is a plausible-looking number, which is what makes this the hardest Week 2 mistake to catch. Check the formula, not the count.

Step 4 — Insert the Histogram Chart

 **Select B6:C16** — labels and counts, headers included. **Not** column A, whose numbers would be charted as a second series.

On the **Insert** tab, find the **Charts** group. Click **Column** → **Clustered Column**.

The chart appears with a gap between bars. Right-click any bar → **Format Data Series** → find the **Gap Width** slider and drag it to **0%**. Double-click the chart title to rename it.

Why Gap Width = 0%?

Histogram bars represent **adjoining ranges on a number line** — nothing sits between the 70–80°F bin and the 80–90°F one, so no visual gap should appear between their bars. A gap turns your histogram into a bar chart, which says the x-axis holds separate categories instead. Bars that touch signal a numeric axis.

 **Not** Insert Statistic Chart → Histogram. That command bins raw values itself, so pointing it at your finished bin table makes it bin your ten counts. It works on raw `Data!B2:B8761` — but then Excel picks the bin edges, not you.

Step 4 — Which Button, and Why It Matters

Two routes to a histogram — they take different input

1 Select the table

	A	B	C
6	Bin low	Bin (°F)	Hours
7	20	20-30	6
8	30	30-40	280
⋮			
16	110	110-120	9

B6:C16 — not column A

2 Insert tab

FileHomeInsertData



Column ▾

2-D Column

Clustered Column

Stacked Column



Statistic Chart ▾

Histogram

Pareto

Box and Whisker

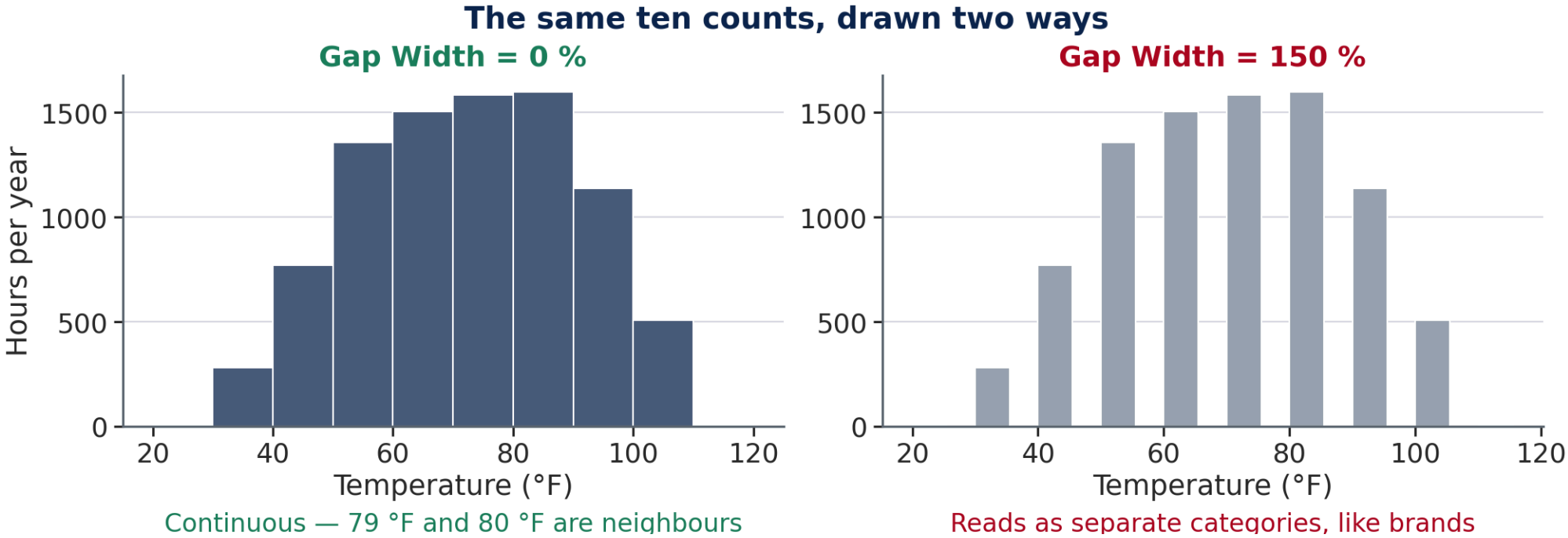
X

✓ Column → Clustered Column, then close the gap to 0% — plots the ten counts you computed.

X Statistic Chart → Histogram on this table bins the counts themselves — you get a histogram of your ten counts.

⚠ That command is not broken — it wants raw Data!B2:B8761, and then it picks the bin edges, not you.

Step 4 — Gap Width, Side by Side



Step 5 — SKEW(): One Number, One Answer

`=SKEW( range )` returns the skewness coefficient — a single number that describes the asymmetry of the distribution.

 **In any blank cell, type:**

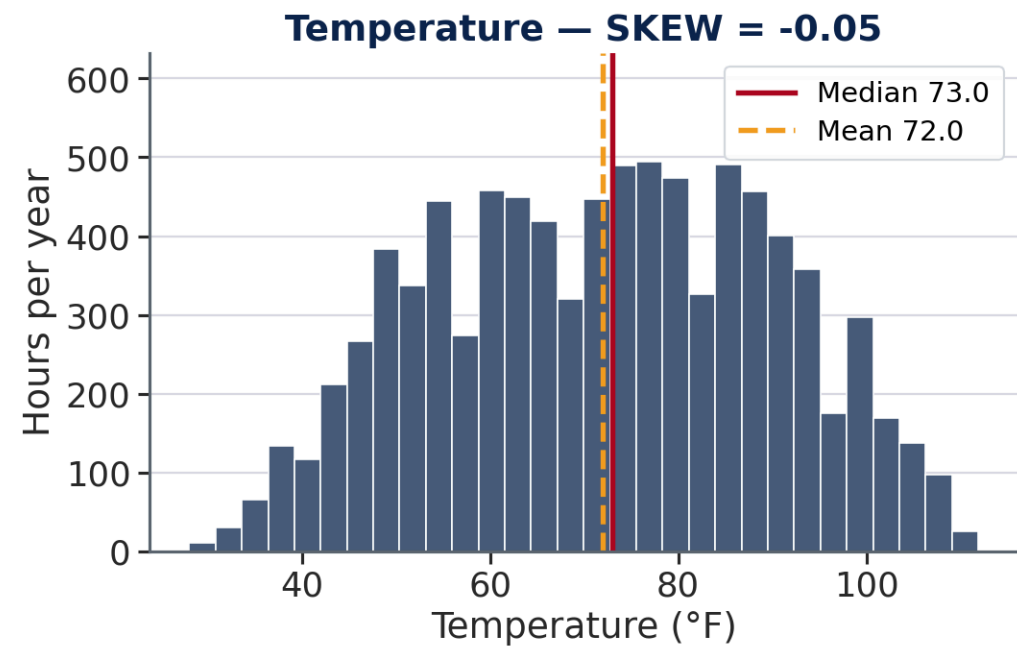
`=SKEW(Data!B2:B8761)` → temperature result ≈ -0.05

`=SKEW(Data!C2:C8761)` → wind speed result $\approx +1.07$

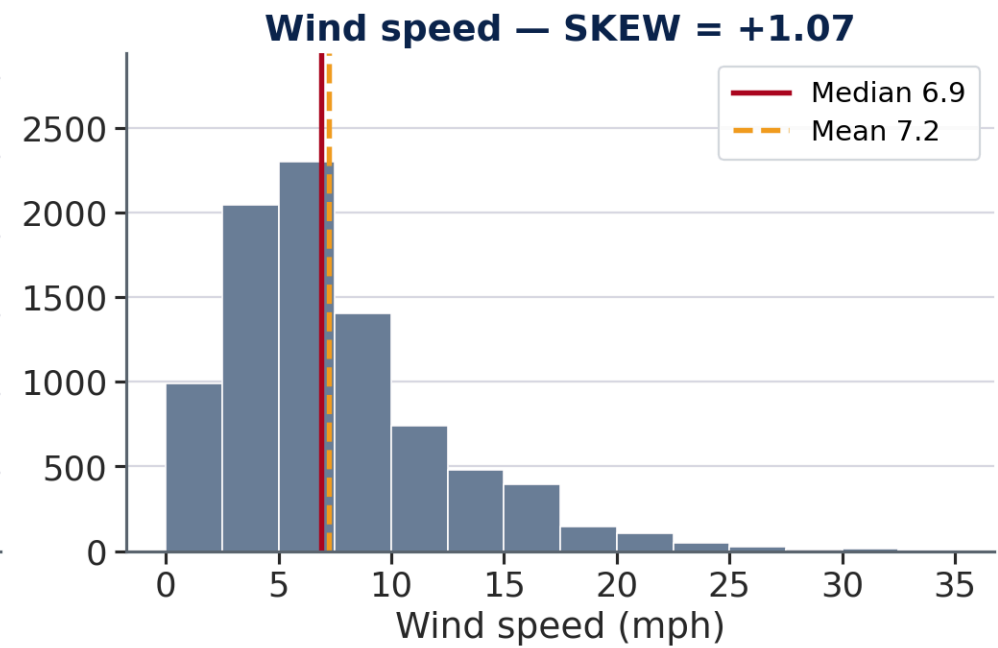
How to read the result:

Value	Meaning	Shape	Tucson example
Near 0 (−0.5 to +0.5)	Roughly symmetric	Mean $\approx$ Median	Temperature: −0.05
Positive ($> +0.5$)	Right-skewed — tail toward high values	Mean $>$ Median	Wind: +1.07
Negative (< -0.5)	Left-skewed — tail toward low values	Mean $<$ Median	—

Step 5 — What -0.05 and $+1.07$ Look Like



Symmetric — mean sits on the median



Right-skewed — the tail drags the mean up

Step 6 — Box-and-Whisker Chart

Excel builds the box plot directly from raw data — no bin table needed.

 **Go to the Data sheet and select B2:B8761** (the entire temperature column, no header)

Insert → Statistical Charts → **Box and Whisker**

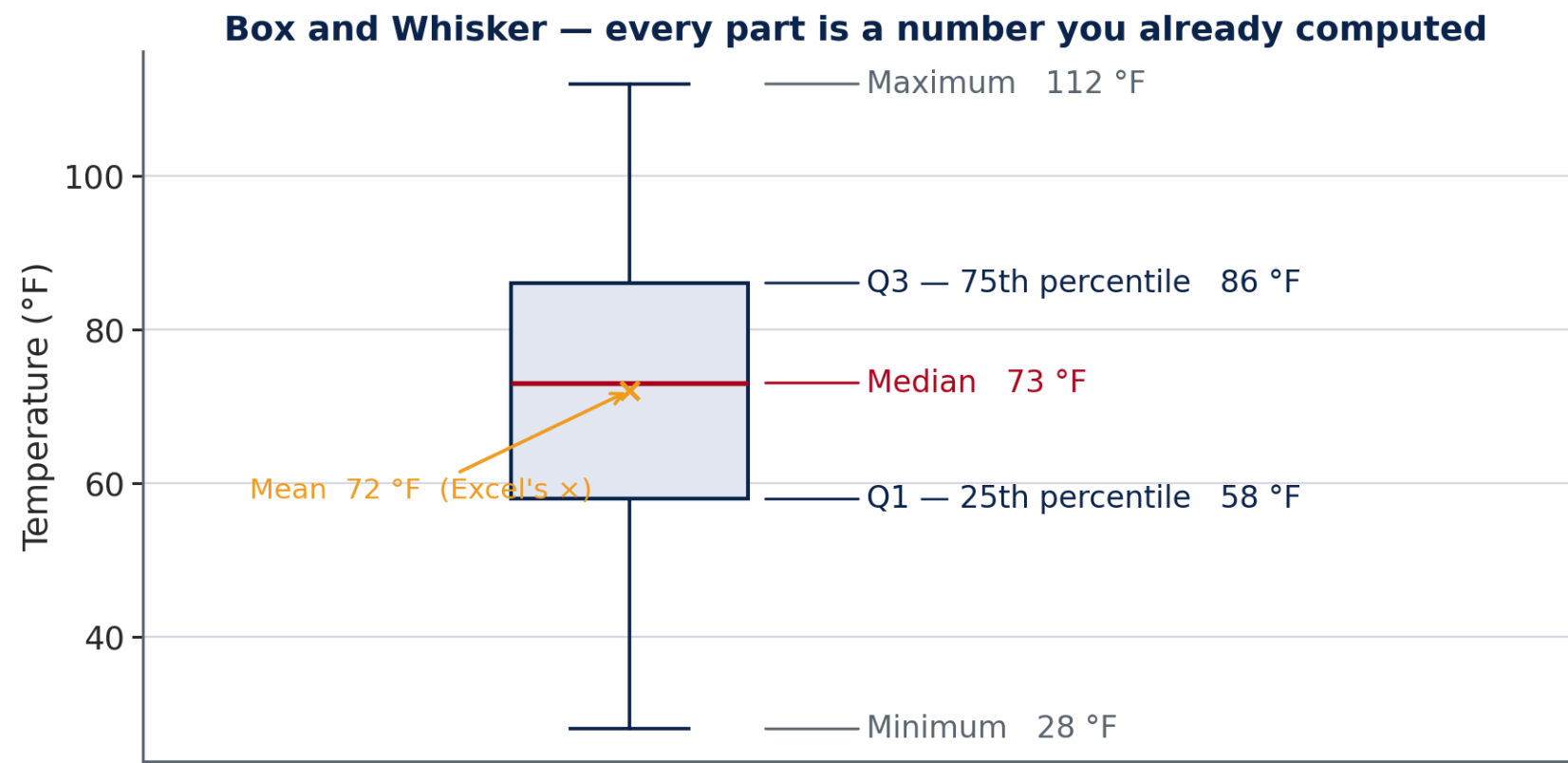
Click Chart Design → Add Chart Element → Axis Titles to label the Y axis "Temperature (°F)". Double-click the chart title to rename it.

Box = middle 50% (Q1→Q3) · **Line** = Median ·
Whiskers = Min/Max · **Dots** = outliers

Reading the result for Tucson temperature:

Part	Value
Bottom whisker	≈ 28°F
Q1 (bottom of box)	≈ 58°F
Median (line)	≈ 73°F
Q3 (top of box)	≈ 86°F
Top whisker	≈ 112°F

Step 6 – Reading the Box



Excel marks the mean with x. It sits just below the median here — that is the -0.05 skew, made visible.

Step 7 – PERCENTILE(): Any Tail Value in One Formula

`=PERCENTILE(range, p)` returns the value below which fraction `p` of the data falls, where `p` is a decimal between 0 and 1.

 **Type these in blank cells and confirm the results:**

Formula	What it finds	Result
<code>=PERCENTILE(Data!B2:B8761, 0.50)</code>	Median (50th percentile)	≈ 73.0°F
<code>=PERCENTILE(Data!B2:B8761, 0.90)</code>	90th percentile temperature	≈ 96.1°F
<code>=PERCENTILE(Data!B2:B8761, 0.95)</code>	95th percentile temperature	≈ 100.0°F
<code>=PERCENTILE(Data!C2:C8761, 0.95)</code>	95th percentile wind speed	≈ 17.2 mph
<code>=PERCENTILE(Data!D2:D8761, 0.90)</code>	90th percentile humidity	≈ 64.1%

Quick Check — Before You Start the Lab

Think, then check

You computed:

- $\text{SKEW}(\text{Temperature}) \approx -0.05$
- $\text{SKEW}(\text{Wind Speed}) \approx +1.07$

Question A: For temperature, will the mean be above or below the median — and by how much?

Question B: For wind speed, which direction is the long tail — toward high values or low values?

What does that mean for sizing a wind-load system?

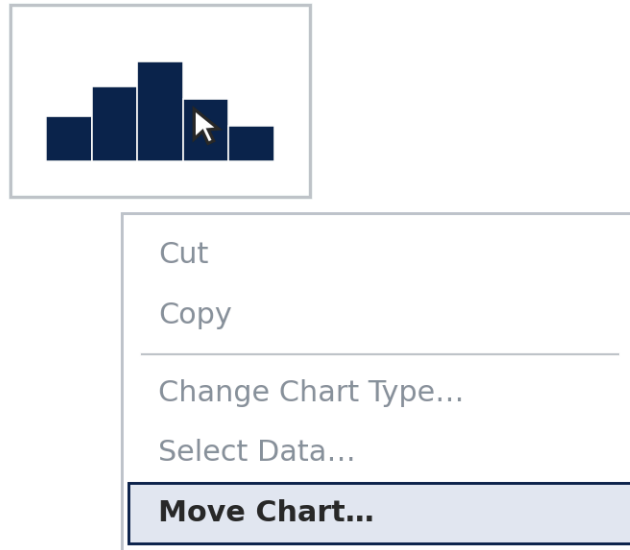
Compare your answers with your neighbor. We'll confirm with the data.

The Step Every Lab Ends With

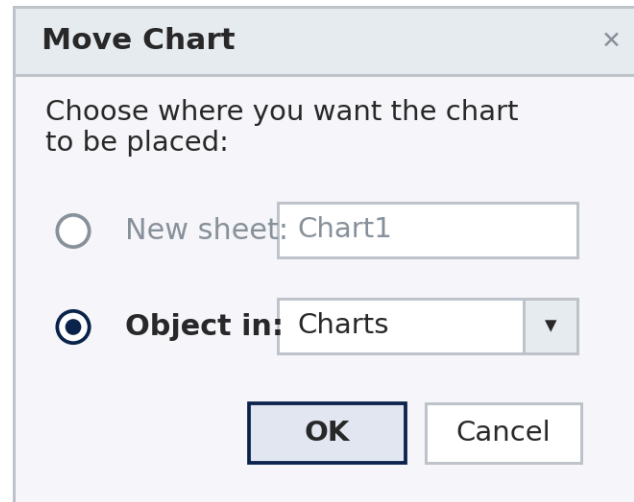
Getting a finished chart into the box it is graded in

Every week. Build the chart on the sheet its data lives on, then move it onto the Charts sheet.

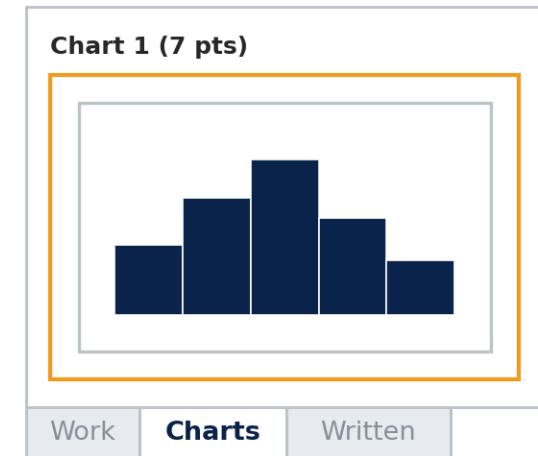
1 Right-click the chart



2 Choose "Object in"



3 It lands in the box



X New sheet — makes a separate sheet called Chart1. The box on Charts stays empty and the points are lost.

✓ Object in → Charts — then drag the chart inside the outlined box.

⚠ Leave it a live chart — a pasted picture of one scores zero.

Quick Reference — Week 2 New Formulas

Task	Formula
Count in bin [lo, hi)	<code>=COUNTIFS(Data!B\$2:B\$8761,">="&A7,Data!B\$2:B\$8761,"<"&(A7+10))</code>
Skewness	<code>=SKEW(Data!B2:B8761)</code>
Nth percentile	<code>=PERCENTILE(Data!B2:B8761, n/100)</code>
Quartile (Q1)	<code>=QUARTILE(Data!B2:B8761, 1)</code>
Variance	<code>=VAR(Data!B2:B8761)</code>

Task	How
Histogram chart	Select <code>B6:C16</code> → Insert → Column → Clustered Column, Gap Width 0%
Box plot	Insert → Statistical Charts → Box and Whisker
Fill bin table down	<code>=COUNTIFS</code> in C7, then double-click the fill handle

⚠ In the bin COUNTIFS, lock the data range with \$ (`Data!B$2:B$8761`) but leave the bin cell (`A7`) unlocked, so it shifts down as you fill.

Before You Submit

⚠ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

1. **File format** — **.xlsx only**. Save → Save As → Excel Workbook (*.xlsx). Files saved as **.xls**, **.csv**, or **.pdf** cannot be graded.
2. **Sheet name** — **Answers**. Do not rename or delete this sheet.
3. **Column layout** — **do not change**. Answers go in Column E. Do not insert, delete, or move columns.
4. **File name** — **CE310_W2_<NetID>.xlsx**, e.g., **CE310_W2_abc123.xlsx**. Fill in the **NetID** cell at the top of the **Answers** sheet as well — that cell is what matches your work to your student record, and a 5-point deduction applies if it is blank. (The weekly assignment is a separate file, named **CE310_W2_Assignment_<NetID>.xlsx**.)
5. **Upload** the **.xlsx** file to D2L Dropbox by **9:00 AM next Wednesday**.

You're Ready — Open the Lab Instructions

Wk02.03_In_Class_Exercise.pdf · Wk02.04_In_Class_Answer_Sheet.xlsx

Your **Data** sheet is built and your setup is complete. The lab picks up from here:

- **Section A** — bin counts for temperature (A1–A3) and wind (A5–A6), the three SKEW values (A4, A7, A8), and CV (A9)
- **Section B** — temperature percentiles (B1–B6), wind percentiles and design pressure (B7–B11), humidity (B12–B13), and the probability bridge (B14)
- **Charts and memo** — the histogram you just built (Chart 1), a temperature box-and-whisker (Chart 2), and the three-paragraph memo

Work independently. Everything not completed in class is take-home.

Raise your hand if your bin counts don't match the expected values on the previous slide — the most common cause is a missing **\$** in the COUNTIFS range.